

NIRANGA NAYANAJITH

DEVELOPER & CYBER SECURITY ENTHUSIAST



0756814599



niranganayanajith195@gmail.com



<https://niranga.vercel.app/>



Ampara 32064, Sri Lanka



<https://www.linkedin.com/in/niranga-nayanajith/>

PROFILE

Undergraduate Developer and Cybersecurity Enthusiast with a focus on system architecture and infrastructure observability. Proven track record in building multi-service backends and automation pipelines using Python, Ballerina, and TypeScript. Specializes in bridging the gap between high-performance software and robust monitoring, leveraging a deep understanding of network security and cloud engineering to ensure system reliability and data integrity. Aims to apply expertise in DevOps and automated workflows to modernize large-scale distributed systems like CernVM-FS.

EDUCATION

Bachelor of Science in Computing and Information Systems

Sabaragamuwa University of Sri Lanka

- Expected Graduation: December 2028

SKILLS

- Software development expertise
- Programming languages: Python, TypeScript, JavaScript, Ballerina, C#, Java
- Web frameworks: Angular, Next.js, Node.js
- Backend services and APIs
- Systems monitoring and observability
- Distributed systems architecture
- Containerization fundamentals
- Linux command-line proficiency
- Version control with Git/GitHub
- Networking and security fundamentals
- Security principles: CIA triad and risk assessment
- Network scanning tools: Nmap
- Application packaging with PyInstaller
- Technical documentation with LaTeX and Jupyter notebooks

CERTIFICATIONS & LICENSES

- Fundamentals of Game Hacking Development – Red Team Leaders
- Cybersecurity Fundamentals – IBM SkillsBuild Student Ambassador Program
- Data Science for Cybersecurity – Red Team Leaders
- Build Your Own AI Chatbot with Microsoft Foundry – Microsoft Learn Student Ambassadors (Sri Lanka)
- AI Buildathon – IEEE Computer Society of KDU
- Networking Basics – Cisco Networking Academy
- SQL Injection – Simplilearn
- Ethical Hacking: Mobile Platforms and Network Architecture – Great Learning
- Introduction to Cyber Security – Great Learning
- Responsive Web Design – freeCodeCamp
- Introduction to Programming Using Python – Kaggle
- Basic in Python – HackerRank
- C Language Gold Level – HackerRank

PROJECTS

1. NailVirtuoso (AI Marketplace & Supreme Admin Console)

Technologies: Next.js, TypeScript, Firebase, Gemini API

- Engineered a “Supreme Admin Console” with real-time API health diagnostics, status dashboards, and Gemini quota monitoring (tracking 1,500 requests/day).
- Implemented a robust Role-Based Access Control (RBAC) system with “Divine Guard” route protection and secure admin-only API endpoints.
- Leveraged Firestore onSnapshot() for live data synchronization across system statistics, inventory, and pending queues.

2. Webify.me (AI-Driven Full-Stack Automation)

Technologies: React, Python (Flask), MongoDB, Gemini API

- Built a complex backend that transforms raw topics into structured content (PPTX), managing communication between a React frontend and a Python-based processing engine.
- Handled large-scale integrations with external APIs (Gemini, Unsplash) and implemented asynchronous task management for content generation

3. 2N Share (Cross-Platform Networking Utility)

Technologies: Python, Flask, Socket, Threading, PyInstaller

- Engineered a multi-threaded backend enabling localized file-sharing with automated local IP discovery for zero-config networking.
- Implemented secure filename handling to prevent directory traversal attacks and ensured data integrity during high-speed transfers.
- Utilized PyInstaller to package optimized, standalone executables for cross-platform reliability.

4. NailDetectAI (Deep Learning Virtual Segmentation)

Technologies: PyTorch, U-Net, Flask, Docker, Gunicorn

- Packaged a deep learning model using Docker and Gunicorn for reproducible and scalable deployment.
- Implemented image segmentation (U-Net) for virtual visualization, demonstrating advanced algorithmic and mathematical problem-solving skills.

5. Voice Assistant & Virtual Mouse (HCI & OS Automation)

Technologies: OpenCV, MediaPipe, PyAutoGUI, Windows APIs

- Solved hardware-software lag using weighted moving average algorithms for smooth real-time interaction with low CPU overhead.
- Integrated Python scripts with low-level Windows APIs for direct system control, showcasing deep OS-level understanding.

MY TECHNICAL INTERESTS

- DevOps and Cloud Infrastructure
- Cybersecurity and Ethical Hacking
- Distributed Systems and Observability
- AI-Driven Automation
- System Performance Optimization